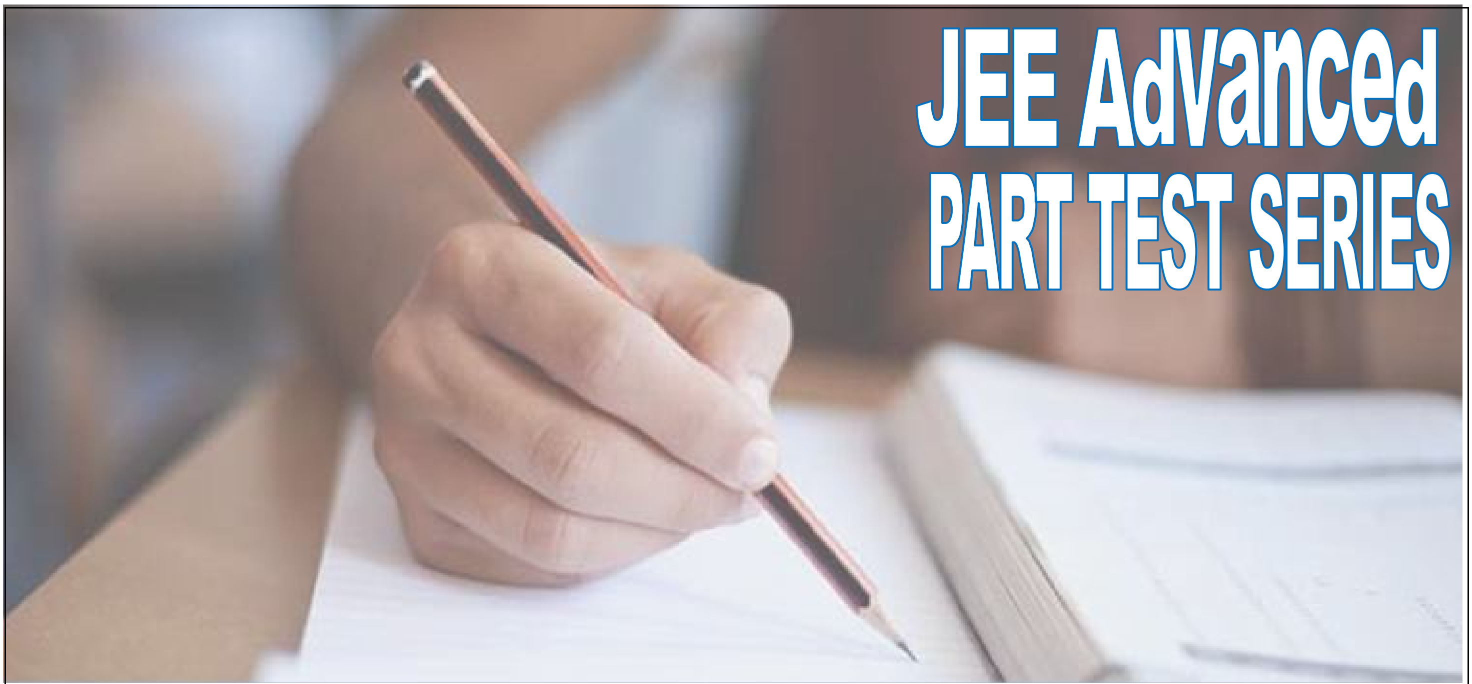

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A right Choice for the Real Aspirant

ICON Central Office , Madhapur – Hyderabad

SEC: Sr's_In Coming	**JEE-ADV(2016-p1)**	**Date: 03-05-20**
Time: 09 A.M TO 12 NOON	**PTA-5**	**Max. Marks:**

SYLLABUS:

Physics : **Viscosity, Surface tension & Elasticity**

Chemistry : **Thermodynamics & Thermo-Chemistry, Isomerism, Reactive intermediates, Electronic factors, Acidity & basicity of organic compounds**

Mathematics : **Matrices & determinants**

Name of the Student: ____________________

H.T. NO: | | | | | | | |

JEE-ADVANCE-2016-P1-Model

Time: 9:00A.M To 12:00 P.M **IMPORTANT INSTRUCTIONS** Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 5)	Questions with Single Answer Type	+3	-1	5	15
Sec – II(Q.N : 6 – 13)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	4	-2	8	32
Sec – II(Q.N : 14-18)	Questions with Integer Answer Type	+3	0	5	15
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 23)	Questions with Single Answer Type	+3	-1	5	15
Sec – II(Q.N : 24 –31)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	4	-2	8	32
Sec – II(Q.N : 32-36)	Questions with Integer Answer Type	+3	0	5	15
Total				**18**	**62**

MATHEMATICS:

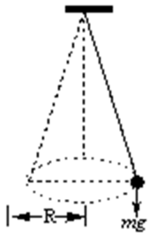
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N:37 – 41)	Questions with Single Answer Type	+3	-1	5	15
Sec – II(Q.N :42–49)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	4	-2	8	32
Sec – II(Q.N : 50-54)	Questions with Integer Answer Type	+3	0	5	15
Total				**18**	**62**

PHYSICS Max Marks: 62

SECTION-I
(SINGLE CORRECT CHOICE TYPE)

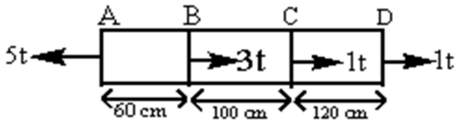
This Section Contains **FIVE** questions, Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of those four options is Correct For each question, darken the bubble corresponding to the correct option in the **ORS**, For each question, marks will be awarded in one of the following categories: Full Marks : +3 If only the bubble corresponding to the correct option in darkened, Zero Marks : 0 If none of the bubbles is darkened. Negative Marks: -1 In all other cases

1 A stone of $0.5kg$ mass is attached to one end of a $0.8m$ long aluminum wire of $0.7\,mm$ diameter and suspended vertically. The stone is now rotated in a horizontal plane at a rate such that wire makes an angle of 85^0 with the vertical. If $Y = 7\times10^{10}\ Nm^{-2}$, $\sin 85^0 = 0.9962$ and $\cos 85^0 = 0.0872$, the increase in length of wire is



A. $1.67\times10^{-3}m$ **B.** $6.17\times10^{-3}m$ **C.** $1.76\times10^{-3}m$ **D.** $7.16\times10^{-3}m$

2 A brass bar having a cross sectional area $10cm^2$ is subjected to axial force as shown in the figure. Take $E = 8\times10^2 t/cm^2$. The total elongation of the bar is

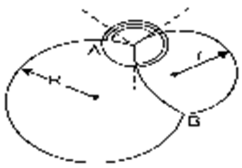


A. + 0.0775 cm **B.** 0.050 cm **C.** –0.0.25 cm **D.** 0.095 cm

3 When water rises in a capillary tube of radius r to height h, then its potential energy U_1, if capillary tube of radius 2r is dipped in same water then potential energy of water is U_2. Then $U_1 : U_2$ will be

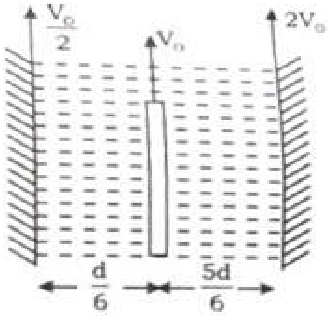
A. 1 : 1 **B.** 1 : 2 **C.** 2 : 1 **D.** 1 : 4

4 A soap - bubble with a radius 'r' is placed on another bubble with a radius R (figure). Angles between the films at the points of contact will be



A. 120^0 **B.** 30^0 **C.** 45^0 **D.** 90^0

5 What is the constant force F required to pull the plate of area A parallel to wall with constant velocity V_0 in the given figure. (η = coefficient of viscosity.)



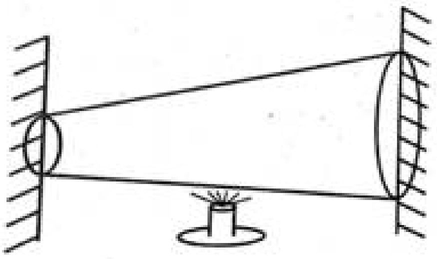
A. $\frac{9\eta AV_0}{2d}$ **B.** $\frac{45\eta AV_0}{8d}$ **C.** $\frac{9\eta AV_0}{5d}$ **D.** $\frac{8\eta AV_0}{3d}$

SECTION-II

(MULTIPLE CORRECT CHOICE TYPE)

This Section contains **EIGHT** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options(s) is (are) correct. For each question, darken the bubble(s) corresponding to all the correct options(s) in the **ORS.** For each question, marks will be awarded in one of the following categories: Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s), is (are) darkened . Partial Marks : +1 For darkening a bubble corresponding to each correct option, Provided NO incorrect option is darkened. Zero Marks : 0 If none of the bubbles is darkened. Negative Marks: -2 In all other cases. , For example, if (A), (C)and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and(B) will result in -2 marks as a wrong option is also darkened.

6 A tapered rod of non uniform cross-sectional area is fitted between two rigid supports. The rod is heated at the middle, then choose the incorrect options.



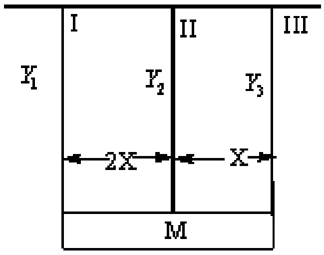
A. Compressive stress is maximum at narrow section

B. Compressive stress is maximum at mid section

C. Compressive stress is maximum at wider section

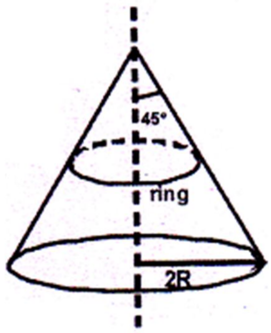
D. Elastic strain developed in the rod is zero as length remains same

7 Three vertical wires I,II and III are supporting a block of mass M in horizontal position. The wires are of equal length and cross-sectional area. It is given that Young's modulus of wire II, is Y_2 and Young's modulus of wire III is Y_3 and it for 1st wire is Y_1. The wires I and III are attached at extreme ends of the block. If $Y_2 = Y_3$ then which of the following are correct [Where T_1, T_2 and T_3 are tension in I, II and III wires]



A. $T_1 = 2T_3$ **B.** $T_2 = T_3$ **C.** $Y_1 = \frac{4Y_2}{3}$ **D.** $T_1 = 2Mg/5$

8 A ring of mass m, radius R, cross sectional area a and Young's modulus Y is kept on a smooth cone of radius 2R and semi vertical angle 45^0, as shown in the figure. Assume that the extension in the ring is small.



A. The tension in the ring will be same throughout

B. The tension in the ring will be independent of the radius of ring if radius of the ring is less than 2R

C. The extension in the ring will be $\frac{mgR}{aY}$

D. Elastic potential energy stored in the ring will be $\frac{m^2g^2R}{8\pi Ya}$

9 When a capillary tube is dipped in a liquid, the liquid surface becomes curved due to force of adhesion (F_1) and force of cohesion (F_2). Choose the correct statement

A. Angle made by resultant force of F_1 and F_2 makes an angle $\tan^{-1}\left(\frac{F_2}{\sqrt{2}F_1 - F_2}\right)$

B. The meniscus will be plane if $F_2 = \sqrt{2}F_1$

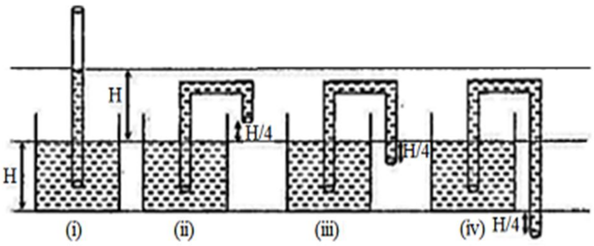
C. $F_2 < \sqrt{2}F_1$, the liquid surface will be concave upward

D. $F_2 > \sqrt{2}F_1$, the liquid will be convex upward

10 If a liquid rises to same height in two similer capillaries of the same material at the same temperature then

A. the weight of liquid in both capillaries must be equal
B. the radius of meniscus must be equal
C. the capillaries must be cylindrical and vertical
D. the hydrostatic pressure at the base of capillaries must be same

11 Water rises to height H in a capillary of radius r and angle of contact is zero as shown in figure. The density and surface Tension of water is ρ and T respectively. Acceleration due to gravity is g and atmospheric pressure is P_0. Now, identical capillary as of (i) is bend in different forms as shown in the figure (ii), (iii) and (iv)



A. radius of curvature of meniscus at the open end of capillary in figure (ii) is $8T / \rho g H$

B. radius of curvature of meniscus at the open end of capillary in figure (iii) is $8T / \rho g H$

C. radius of curvature of meniscus at the open end of capillary in figure (iv) is $8T / \rho g H$

D. liquid will be flowing out in figure (iv)

12 A spherical solid body is dropped inside a vast expanse of viscous liquid of large depth and of coefficient of viscosity η. The density of the solid is greater than that of the liquid. The time taken by the body to attain the 90% of the steady state velocity is dependent on

A. density of the liquid **B.** density of the solid

C. diameter of the sphere **D.** coefficient of viscosity

13 Two balls of same material (density ρ) but radii r_1 and r_2 are joined by a light long inextensible vertical thread and released from a large height with string in slack condition in a medium of co-efficient of viscosity η. Neglect buoyancy and assume acceleration due to gravity to be a uniform. Then [assume $r_1 > r_2$]

A. The terminal velocity acquired by the balls is $\frac{2}{9}\frac{\rho g}{\eta}\left[r_1^2 - r_1 r_2 + r_2^2\right]$

B. Tension in the string when balls are moving with terminal velocity is $\frac{4}{3}\pi\rho g\left[r_1^2 r_2 - r_2^2 r_1\right]$

C. after the release, both ball will have same acceleration

D. the string will get taut some time after the release

SECTION 3
(INTEGER ANSWER TYPE)

This section contains **FIVE** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive For each question, darken the bubble corresponding to the correct integer in the **ORS.** For each question ,marks will be awarded in one of the following categories :
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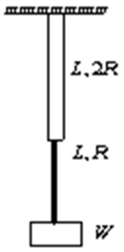
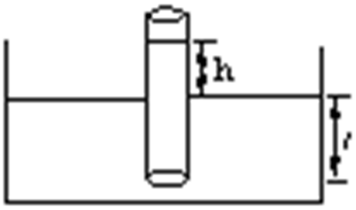
14 Two wires of the same material (Young's modulus Y) and same length L but radii R and 2R respectively are joined end to end and a weight W is suspended from the combination as shown in the figure. The elastic potential energy in the system is $\frac{NW^2L}{8\pi R^2 Y}$ where N=



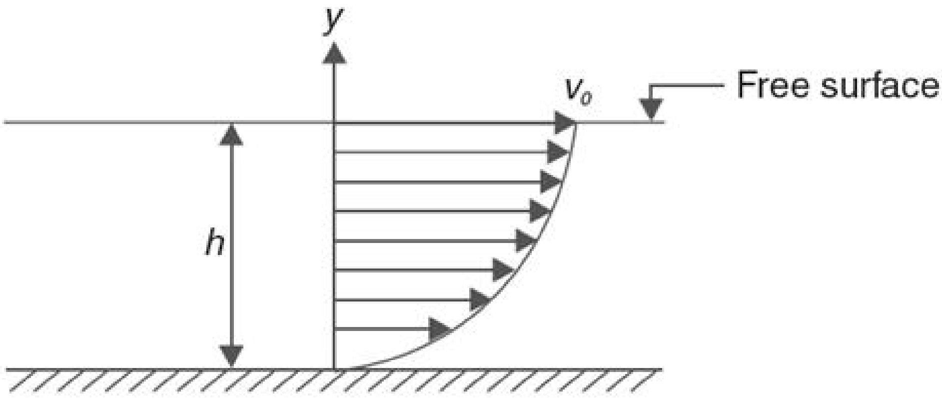
15 A load of 981 N is suspended from a steel wire of radius 1mm. The maximum angle through which the wire with a load can be deflected so that it dos not break when the load passes through the position of equilibrium is computed as $\cos^{-1}\left[\alpha(0.081)\right]$.
Find α. (Breaking stress is $7.85\times10^8\,N/m^2$)

16 A cylinder with a movable piston contains air under a pressure P_1 and a soap bubble of radius 'r'. The pressure P_2 to which the air should be compressed by slowly pushing the piston into the cylinder for the soap bubble to reduce its size by half will be :n x $\left[P_1 + \frac{3S}{r}\right]$, where S is the surface tension, where n=______

17 A capillary tube is dipped in water to depth l and the water rise to a height $h(<1)$ in the capacity tube. The lower end of the tube is closed in water by putting a thumb over it. The tube is now taken out and the thumb is removed from the lower end and it is kept open. The length of the liquid column in the tube will be x times of h. Find x



18 A liquid is flowing through a horizontal channel. The speed of flow (v) depends on height (y) from the floor as $v = v_0\left[2\left(\frac{y}{h}\right) - \left(\frac{y}{h}\right)^2\right]$. Where h is the height of liquid in the channel and v_0 is the speed of the top layer. Coefficient of viscosity is η. Calculate the shear stress that the liquid exerts on the floor. Answer in multiples of $\frac{\eta v_0}{h}$



CHEMISTRY

Max Marks: 62

SECTION-I
(SINGLE CORRECT CHOICE TYPE)

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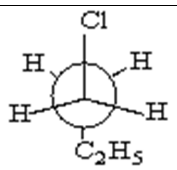
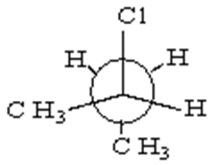
19  and  are a pair of

A. Conformational isomers

B. Chain isomers

C. Positional isomers

D. Identical Compounds

20 The number of isomers from the compound with molecular formula $C_2BrClFl$ is

A. 3 **B.** 4 **C.** 5 **D.** 6

21 An ideal gas expands in volume from $1\times10^{-3}m^3$ to $1\times10^{-2}m^3$ at 300 K against a constant pressure a $1\times10^5 Nm^{-1}$. The work done is

A. $-900\ J$ **B.** $900KJ$ **C.** $270KJ$ **D.** $-900KJ$

22 The heat of neutralization of NaOH with HCl is 57.3 KJ and with HCN is 12.1 KJ. The heat of ionization of HCN is

A. +69.4 KJ **B.** +45.2 KJ **C.** - 45.2 KJ **D.** -69.4 KJ

23 $A \rightleftharpoons B,\ K=8......(I)$

$B \rightleftharpoons C,\ K=10......(II)$

$C \rightleftharpoons D,\ K=0.01......(III)$, then correct order of ΔG^0 values of processes at the same temperature is

A. III > I > II **B.** III > I = II **C.** I > III > II **D.** II > I > III

SECTION-II

(MULTIPLE CORRECT CHOICE TYPE)

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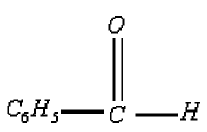
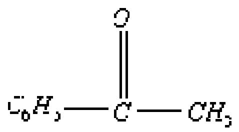
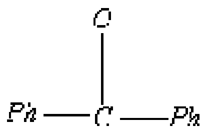
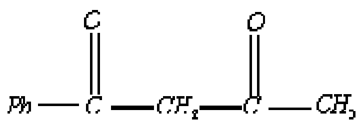
24 Keto-enol tautomerism is observed in :

A.  **B.**  **C.**  **D.** 

25 The specific heats of a gas at constant volume and constant pressure are 0.038 cal $\deg^{-1} g^{-1}$ and 0.062 cal $\deg^{-1} g^{-1}$.

A. The atomicity of the gas molecule is 2.

B. The molar of the gas is 83 g mol^{-1}.

C. C_P for the gas is 5 cal $\deg^{-1} mol^{-1}$.

D. The gas has such a low critical temperature that it cannot be condensed to form a liquid above 10 k.

26 Which is/are correct among the followings ?

A. $\left(\frac{\partial H}{\partial T}\right)_P = C_P$ **B.** $\left(\frac{\partial \Delta H}{\partial T}\right)_P = \Delta C_P$ **C.** $\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_P dT$ **D.** $\left(\frac{\partial U}{\partial T}\right)_V = \Delta C_P$

27 **Passage-2**

Heat of atomisation is the quantity of heat required to produce one mole of atoms in the gaseous state from the element in the standard state is called the heat of atomisation of the element. The heats of atomisation of the elements are determined from spectroscopic measurements and some thermal data.

Answer the following questions based on the paragraph.

A) $C_{(gr)} + 2H_{2(g)} \rightarrow CH_4;\quad \Delta H_f^\circ = -17.9\ Kcal$

B) $C_{(s)} \rightarrow C_{(g)};\quad \Delta H_1^\circ = 170.9\ Kcal$

C) $2H_{2(g)} \rightarrow 4H_{(g)};\quad \Delta H_2^\circ = 4 \times 52.1\ Kcal$

Q : Given are

1) $2C_{(s)} + 2H_{2(g)} \rightarrow C_2H_{4(g)}; \quad \Delta H_f^\circ = 12.5 \; Kcal$

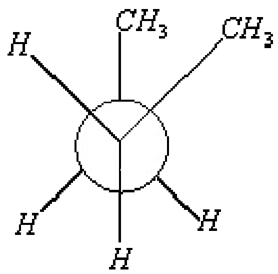
2) $2C_{(s)} \rightarrow 2C_{(g)}; \quad \Delta H^\circ = 2 \times 170.9 \; Kcal$

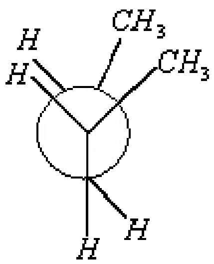
3) $2H_{2(g)} \rightarrow 4H_{(g)}; \quad \Delta H^\circ = 4 \times 52.1 \; Kcal$

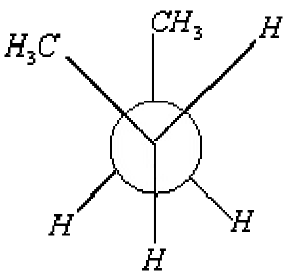
The energy of C-H bond = 99 $Kcal$ What is the bond energy of $C = C$ bond?

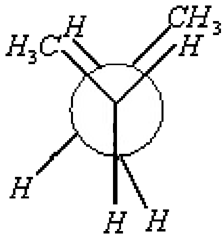
A. 141 $Kcal$ (approx)
B. $(537.7 - 396) \; Kcal$
C. $[537.7 - 4 \times BDE_{C-H}]Kcal$
D. none of the these

28 Which of the following represent staggered conformation of n-butane?

A. 

B. 

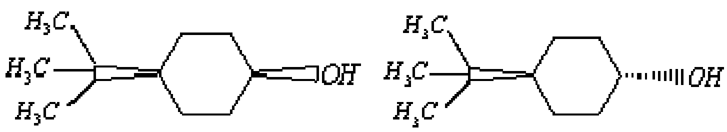
C. 

D. 

29 Which of the following compounds are optically inactive due to internal compensation?

A. 2,3-pentanediol
B. 1,2-propanediol
C. 2,3-butanediol
D. 2,3,4-bromoheptane

30 Which of the following are correct regarding these molecule.



A. Both compounds contain plane of symmetry

B. Both are enantiomers

C. Both are diastereomers of each other

D. Both are homomers

31 $C(S)+H_2O(g)\rightarrow CO(g)+H_2(g)$;

$C(S)+H_2O(g)\rightarrow CO(g)+H_2(g)$; Mark out the correct statement(S)

A. Reaction is spontaneous even at the room temperature.

B. Reaction is not spontaneous even at the room temperature.

C. Reaction is spontaneous above $705°C$

D. ΔH outweighs the entropy factor $T\Delta S$ at the room temperature.

SECTION 3
(INTEGER ANSWER TYPE)

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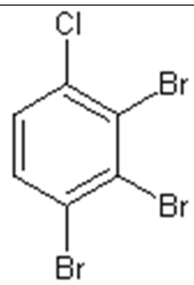
32 Number of positional isomers for



is........?
(Excluding the given structure)

33 What is the active mass of an ideal gas at 44.8 atm pressure and 0^0C temperature in $mol.lt^{-1}$

34 The maximum work done in expanding 16 gm of oxygen at 300K and Occupying a volume of 5 dm^3 isothermally until the volume become 25 dm^3 is $-2.01\times10^{x}J$ value of x is

35 For a reversible reaction , $A \rightleftharpoons B$, the equilibrium constant is expressed as $\log_{10}k=0.47-\dfrac{2000}{T}$ (all values in S.I units). The standard entropy change (ΔS^0) of the reaction is very nearer to ______(in $Jk^{-1}\ unit$)

36 Among the following , for many substances $\Delta G_f^{\circ}=O$:
Liquid water, black phosphorous, graphite, solid $NaCl$, Hydrogen gas, Mercury vapour, hydrated copper sulphate, solid iron.

MATHEMATICS

Max Marks: 62

SECTION-I
(SINGLE CORRECT CHOICE TYPE)

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37 $A=\begin{bmatrix} a & b \\ b & -a \end{bmatrix}$ and $\mathrm{MA} = \mathrm{A}^{2m},\ m \in N$ for some matrix M, then which one of the following is correct?

A. $M=\begin{bmatrix} a^{2m} & b^{2m} \\ b^{2m} & -a^{2m} \end{bmatrix}$ **B.** $M=\left(a^2+b^2\right)^m\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

C. $M=\left(a^m+b^m\right)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ **D.** $M=\left(a^2+b^2\right)^{m-1}\begin{bmatrix} a & b \\ b & -a \end{bmatrix}$

38 The product of the matrices $A=\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$ and $A=\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$ is a null matrix if $\theta-\phi=$

A. $(2n+1)\frac{\pi}{2}$ **B.** $n\pi$ **C.** $2n\pi$ **D.** $n\frac{\pi}{2}$

39 $(x_1,y_1),(x_2,y_2)$ *and* (x_3,y_3) are the vertices of a triangle. Then the equation

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}+\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}=0$$ represents

A. Altitude through A **B.** Median through A

C. Equation of the $\perp^{lar}$ bisector of BC **D.** Angular bisector of A

40 The system of equations $x+y+z=6,\ x+2y+3z=10,\ x+2y+\lambda z=k$ is inconsistent if $\lambda=-,\ k\neq -$

A. 3 , 7 **B.** 3, 10 **C.** 7,10 **D.** 10,3

41 **Paragraph: III**

Consider the determinant $\Delta=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$

$M_{ij}=$ Minor of the element of i^{th} row and i^{th} column

$C_{ij}=$ Cofactor of the element of i^{th} row and i^{th} column

Q. If all the elements of the determinant are multiplied by 2, then the value of the new determinant is

A. 0 **B.** 8Δ **C.** 2Δ **D.** $2^9 . \Delta$

SECTION-II

(MULTIPLE CORRECT CHOICE TYPE)

This Section contains **EIGHT** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct. For each question, darken the bubble(s) corresponding to all the correct option(s) in the **ORS.** For each question, marks will be awarded in one of the following categories: Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s), is (are) darkened. Partial Marks : +1 For darkening a bubble corresponding to each correct option, Provided NO incorrect option is darkened. Zero Marks : 0 If none of the bubbles is darkened. Negative Marks: -2 In all other cases., For example, if (A), (C)and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and(B) will result in -2 marks as a wrong option is also darkened.

42 If $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$ then

A. $x=1$ **B.** $x=2$ **C.** $x=3$ **D.** None of these

43 If the sum of two idempotent matrices is idempotent then

A. $AB+BA=O$ **B.** $AB=BA=I$ **C.** $AB=BA=O$ **D.** $AB \neq BA$

44 Let A, B and C be 2×2 matrices with entries from the set of real numbers, Define 0 as follows $AoB=\frac{1}{2}(AB+BA)$, then

A. $AoB=BoA$

B. $AoI = A$, I is an identity matrix of order2

C. $AoA=I^2$

D. $Ao(B+C)=AoB+AoC$

45 If D_1 *and* D_2 are two 3×3 diagonal matrices, then

A. D_1D_2 is a diagonal matrix **B.** $D_1D_2=D_2D_1$

C. $D_1^2+D_2^2$ is a diagonal matrix **D.** None of these

46 $\begin{bmatrix} 1 & -2 & 3 \\ 2 & -1 & 4 \\ 3 & 4 & 1 \end{bmatrix}$ is a

A. Rectangular matrix **B.** Singular matrix

C. Square matrix **D.** Non singular matrix

47 If $g(x)=\begin{vmatrix} a^{-x} & e^{x\log_e a} & x^2 \\ a^{-3x} & e^{3x\log_e a} & x^4 \\ a^{-5x} & e^{5x\log_e a} & 1 \end{vmatrix}$, then

A. $g(x)+g(-x)=0$ **B.** $g(x)-g(-x)=0$ **C.** $g(x)\times g(-x)=0$ **D.** None of these

48 If $AB = A$ and $BA = B$, then

A. $A^2B = A^2$ **B.** $B^2A = B^2$ **C.** $ABA = A$ **D.** $BAB = B$

49 The determinant $\begin{vmatrix} a & b & \alpha x+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$ is equal to 0, if

A. a,b,c are in A.P. **B.** a,b,c are in G.P.

C. a,b,c are in H.P. **D.** $(x-\alpha)$ is a factor of $ax^2+2bx+c$

SECTION 3
(INTEGER ANSWER TYPE)

This section contains **FIVE** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive For each question, darken the bubble corresponding to the correct integer in the **ORS.** For each question ,marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened. Zero Marks : 0 In all other cases

50 $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ where a, b, c are real positive numbers such that $abc=1$ and $AA^T = I$ then $a^3+b^3+c^3 =$

51 If $A = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} = B^3 + C^3$ where B and C are 2×2 matrices with integer elements then $-trace\,B - trace\,C$ must be equal to ....

52 If x, y, z are non-zero real numbers and $\begin{vmatrix} 1+x & 1 & 1 \\ 1+y & 1+2y & 1 \\ 1+z & 1+z & 1+3z \end{vmatrix} = 0$ then $-\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ is equal ____

53 Let a, b, c are real numbers such that $a+b+c$ is positive
$A = bc-a^2, B = ac-b^2, C = ab-c^2$ also $\begin{vmatrix} A & B & C \\ B & C & A \\ C & A & B \end{vmatrix} = 81$ then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} =$

54 The number of all possible values of θ, where $0<\theta<\pi$,for which the system of equations $(y+z)\cos 3\theta = (xyz)\sin 3\theta$, $x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$,
$(xyz)\sin 3\theta = (y+2z)\cos 3\theta + y\sin 3\theta$ have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$,is